Individual Reflection Report

AI and Text Analytics Coursework – Spring 2026
University of Bristol
Team 16, Task 1: PubMedQA Biomedical Question Answering

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1 Contributions Summary

I led the **text-representation axis** (Axis 1), investigating how input encoding affects biomedical reasoning performance. I implemented an incremental strategy from sparse baselines to full fine-tuning, developing a custom stratified splitting script (`data_splitter.py`) to ensure experimental rigour across the team.

My concrete contributions are:

- **TF-IDF + LR Baselines (A1a/b/c):** Implemented three input variants (Q, Q+Ctx, Q+Ctx+Ans[1]) to establish sparse performance ceilings (`baseline_lr_group.py`).
- **BioBERT Embeddings + LR (A1d):** Extracted frozen representations from `biobert-v1.2` to evaluate contextual embeddings for linear classification (`modeling_biobert_lr.py`).
- **BioBERT Fine-tuning (A1e/f):** Adapted the full model with weighted cross-entropy for Q+Ctx and oracle Q+Ctx+Ans inputs (`modeling_biobert_ft.py`).

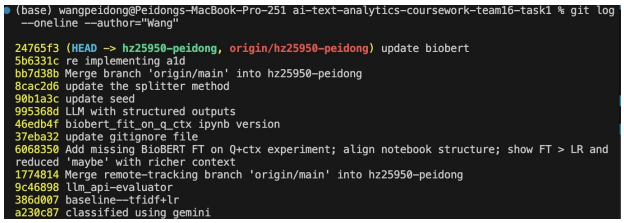

Figure 1: *GitHub commit activity – commits attributed to Peidong Wang.*

[1] **Ans** (long-answer) is a human-written reasoning passage provided in PubMedQA. As it would be unavailable during inference, Q+Ctx+Ans acts as an **oracle** configuration to probe the model's upper-bound capacity.

2 Technical Understanding

2a. Technical Challenge: Context Window Optimization (A1d)

A significant challenge was that frozen BioBERT embeddings (A1d) initially underperformed sparse baselines. I diagnosed this as a **context window limitation**: the default sentence-transformer configuration truncated inputs at 100 tokens, losing critical evidence. I re-implemented the extraction pipeline using `AutoModel` with a 384-token window and mean-pooling. This optimization raised test macro-F1 from 0.353 to 0.419 (+18.7%), demonstrating that representation quality was limited by encoding parameters rather than model expressiveness. Additionally, I implemented weighted cross-entropy to handle the extreme class imbalance (11% *maybe* class, all results are recorded in `AITA_Biobert_fit_on_q_ctx.ipynb`).

2b. Impact on Group Results

My Axis-1 experiments provided the team's central empirical finding: while sparse models plateau regardless of input richness, transformer models can leverage extended context (A1f), achieving 0.466 macro-F1. This established the "representation matters" narrative and motivated our final transition to PubMedBERT (A1g), which yielded our peak result of 0.604. My work ensured that the baseline and intermediate transformer experiments were statistically robust and properly weighted.

2c. Reflection and Future Work

Exploratory zero-shot testing with the Gemini API achieved 0.611 macro-F1, surpassing our fine-tuned models. This indicates that future work should ex-

plore generative reasoning (e.g., Chain-of-Thought) to improve evidence detection for the *maybe* class, moving beyond the representation-focused scope of Axis 1.

3 Teamwork

The team coordinated via **GitHub**, **WhatsApp**, and **Jira**, with weekly **Microsoft Teams** meetings to resolve blockers. To ensure inclusion and consistency, all experimental results were cross-checked by teammates.

Role assignments were as follows:

- **Peidong Wang** – I led the **text-representation axis (Axis 1)**, developing the TF-IDF baselines and BioBERT fine-tuning pipelines. My focus was on optimizing input composition (e.g., adding long-answer channels), tuning the encoding context window.
- **Abdullah** – Data loading, preprocessing, and 10-fold CV infrastructure.
- **Zhengjie** – Axis-2 classical models and ensemble methods.
- **Abhigyan** – Error analysis and misclassification inspection.
- **Tanishk** – Advanced transformer models (PubMedBERT) and final integration.

A shared random seed (`SEED=42`) and a unified data-splitting protocol were adopted by all members to guarantee reproducibility.